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Piano accompaniment: exploring the role of information dynamics, communication practices and cultural influence

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This study investigates the multidimensional nature of piano accompaniment in collaborative music making, focusing on musicians lived experiences, multidirectional information flow, communication strategies, and their cultural implications. While prior research has often examined technical and pedagogical aspects of accompaniment separately, limited attention has been given to how these dimensions intersect in practice, particularly within hybrid cultural contexts such as contemporary China. To address this gap, a qualitative research design was adopted. Data were collected through in-depth semi-structured interviews with 21 participants, including professional piano accompanists, music educators, and music students from different regions of China. The interview data were systematically analyzed using qualitative analysis software to identify recurring patterns and strategies related to technical execution, emotional coordination, and cultural negotiation in accompaniment practice. The findings reveal four interrelated themes that characterize the experience of collaborative accompaniment. First, accompaniment is experienced as a relational and adaptive process, shaped by emotional resonance and sensitivity to interpersonal dynamics. Second, the quality of information accessed during preparation and performance whether derived from digital resources or human mentorship plays a decisive role in rehearsal efficiency and performance outcomes. Third, effective communication emerges through a dynamic interplay of verbal cues and spontaneous gestural interaction rather than through verbal instruction alone. Finally, cross-cultural literacy is shown to be essential for achieving artistic coherence, particularly in educational and performance contexts that integrate Western art music with traditional Chinese musical practices. By integrating perspectives from collaboration, information quality, and cultural-historical activity frameworks, this study conceptualizes piano accompaniment as a multifaceted practice that is simultaneously technical, communicative, and cultural. The findings offer practical and theoretical insights for music education and professional training, highlighting communication fluency and cultural competence as core skills in collaborative performance settings.

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Introduction

In the increasingly complex landscape of collaborative music-making, piano accompaniment serves as both an artistic and communicative bridge, integrating technical expertise with interpretive sensitivity. A single glance, a subtle shift in timing, or a resonant chord progression can transform a solo performance into a dialogic musical experience. Despite its essential role, piano accompaniment is often overshadowed by the prominence of soloists, and its dynamics remain underexplored, particularly in nuanced and cross-cultural contexts.

Traditionally viewed as a supportive function, the accompanist's role has evolved into one of interpretive co-creation. Recent studies emphasize the significance of communication, adaptability, and a shared artistic vision in ensemble work (Jordanous and Smaill 2009; Kubota 2009). Furthermore, technological advancements such as AI-assisted rehearsal tools and digital performance analytics are reshaping how accompanists prepare, interpret, and engage with music (Solomonova et al. 2023). These developments necessitate a re-examination of accompaniment not only as a technical craft but as a multifaceted interaction shaped by information flow, communication strategies, and cultural context. In this context, *information dynamics* refers to the fluid processes through which musical knowledge, interpretative cues, and performance intentions are exchanged, interpreted, and refined among collaborators. This includes both explicit verbal instructions and tacit nonverbal signals, as well as the mediation of these exchanges through digital technologies and institutional norms. Understanding how such informational flows evolve and influence collaborative decision-making is essential for analyzing the complexity of ensemble performance.

While the literature has thoroughly addressed the pedagogical and technical dimensions of accompaniment (Rieger and Manzey 2022), less attention has been paid to the experiential realities of accompanists, their affective labor, decision-making under aesthetic pressure, and negotiation of information across human and digital domains (Zhang and Wang 2024). Moreover, the cultural dimension of accompaniment, particularly in hybrid settings such as China, where Western classical and indigenous traditions intersect, remains empirically understudied (Rahman et al. 2024; Wang et al. 2021). These gaps are particularly pressing in globalized contexts where musicians must navigate linguistic, technological, and cultural pluralism in real-time collaboration.

To address these gaps, this study adopts a qualitative approach to explore how piano accompanists and their collaborators in China experience and negotiate four core dimensions: (1) musical experiences, (2) information quality and relevance, (3) communication practices, and (4) cultural influences. The study is grounded in Collaboration Theory, Information Quality Theory, and Cultural-Historical Activity Theory (CHAT), offering a theoretical framework for analyzing how interpersonal, informational, and socio-cultural dynamics shape performance outcomes.

Ultimately, this study seeks to contribute to the literature by offering a context-sensitive and practice-informed understanding of how musicians collaboratively construct meaning in piano accompaniment. In the spotlight of accompanists and their players, the relationship between human beings in music is understood not as the product of technique and pedagogy but through emotion, communication, and culture. The implications of the findings for music education, ensemble training, and cross-cultural performance models are discussed.

This study draws upon three interrelated theoretical frameworks. Collaboration Theory emphasizes the role of shared goals, mutual responsiveness, and coordinated action in collective tasks, offering a lens through which musical co-creation can be understood.

Information Quality Theory concerns the accuracy, relevance, and contextual richness of information exchanged in decision-making processes essential for understanding how musicians evaluate and synthesize interpretive input. Cultural-Historical Activity Theory (CHAT) highlights the embeddedness of human action within sociocultural contexts, focusing on how historical traditions, institutional norms, and individual agency interact in shaping artistic practice. Together, these frameworks enable a multi-dimensional analysis of accompaniment that integrates technical, communicative, and cultural elements.

Literature review

Musical collaboration and information dynamics. The juncture between musical performance and information transfer has become a significant focus of research. Studies have found that successful accompanists and soloists interact based on shared interpretative intent and on-the-fly responses in collaborative tasks (Springer and Silvey 2022). Mutual synchronization, which extends beyond mere technical synchronization, leverages an expressive coherence rooted in auditory and visual feedback pathways (Venkatesh et al. 2023). During preparation, information dynamism also includes score interpretation, stylistic negotiation, and rehearsal discourse, confirming the accompanist's dialogic (not just accompanying) nature.

Studies have confirmed that continuous feedback, learning-based CS, and the formation of a psychological contract were the critical factors in successful KS (Waseem Bari et al. 2021). These findings are consistent with our participants' experiences regarding the association between the use of AI tools and human mentorship, as well as the increased accuracy and clinical utility of interpretative decisions. Structured rehearsal formations serve as cognitive scaffolding, facilitating a balance between algorithmic precision and human musical intuition.

Technological advancements have further expanded this area, as technologies such as motion capture and sensor-based methods aid in assessing physical gestures and interactional timing. Machine learning models are also being used to account for ensemble response patterns, such as how musicians handle the interpretative load, ensemble synchronization, and temporal accuracy. Herein, these have been shown to favor a many-dimensional and cognitively complex explanation of piano accompaniment activity, which is worthy of theoretical and pedagogical consideration (John et al. 2024).

Pedagogical and emotional dimensions of accompaniment. The experiential realities of musicians in accompaniment settings reflect a complex balance between technical competence and emotional communication. Studies show that successful collaboration requires shared goals, trust, and adaptable artistry (Nugrahu et al. 2023). Frequent challenges include tempo mismatches, ambiguous cues, and interpretative conflicts, all of which can compromise the integrity of performance. However, when effective communication frameworks, such as structured feedback or rehearsal debriefs, are implemented, musicians report enhanced clarity and relational cohesion (Bussu and Mangiarulo 2024).

Pedagogically, accompaniment fosters ensemble sensitivity and artistic growth. Beginner musicians benefit from harmonic support, while advanced students use accompaniment as a medium for interpretative negotiation and collaborative refinement (Bussu and Mangiarulo 2024). These relational and affective aspects reinforce the role of accompaniment in nurturing musical agency and interdependence.

Information quality and technology use. Information quality significantly affects musical collaboration. Empirical evidence suggests that musicians rely on a variety of information sources, including teachers, peers, and, increasingly, AI systems, for interpretative guidance (Adhikari 2023; Raza 2025). While educators offer experiential contextualization, AI tools assist in pattern recognition and performance prediction (Atanacković 2024). Nonetheless, scholars emphasize that algorithmic input often lacks the emotional nuance and contextual depth that human mentors provide (Ben-Tal et al. 2021; Lee et al. 2025).

High-quality information contributes to ensemble trust and lowers uncertainty. Integrating digital tools in rehearsal enhances technical precision, but human oversight remains essential to contextualize insights (Li 2025). The collective use of validated, emotionally resonant, and context-sensitive information supports confidence in rehearsal decisions and contributes to cohesive performance outcomes.

Communication practices in ensemble settings. Studies of communication among musicians underscore the importance of both verbal and nonverbal modalities. Verbal exchanges during rehearsal clarify roles and resolve conflicts, while nonverbal cues, including facial expressions, gestures, and eye contact, facilitate real-time synchrony during the performance (Communication 1972; Davis 2023; Puerto and Thue 2017). These practices form a communication ecosystem that supports interpretative alignment and mutual responsiveness (Hall et al. 2021).

Challenges such as unequal power dynamics or vocabulary limitations can hinder collaboration (Atika 2024). However, when musicians openly discuss artistic goals and engage in reflective dialogue, they develop relational trust and ensemble cohesion (Seddon 2005). These practices are essential in establishing professional norms and fostering mutual respect in performance contexts.

Cultural contexts and interpretative practices. Cultural frameworks significantly shape accompaniment practices, influencing stylistic decisions, performance expectations, and interpersonal dynamics. Research in cross-cultural contexts, particularly China, illustrates how musicians synthesize Western classical traditions with indigenous aesthetics, navigating hybrid repertoires and interpretative standards (Clayton 2016; Trehub et al. 2015).

These interactions require cultural sensitivity, as musicians align phrasing, rhythm, and ornamentation with localized expectations (Saarikallio 2012; Zhan 2025). Understanding cultural heritage strengthens emotional engagement and enhances ensemble cohesion, particularly when collaborators share perspectives and co-create interpretations across diverse backgrounds (Crane 2016; Wu 2023).

Theoretical framework. This study is anchored in two complementary frameworks: Social Exchange Theory and CHAT. Social Exchange Theory posits that collaboration is sustained by reciprocal exchanges of resources, including emotional, informational, and artistic (Foot 2014; Wilson 2014). In the course of accompaniment, this can be viewed as a process of trust, wherein the interaction of aural and visual cues, as well as co-regulated expression, shapes the evolution of trust (Cong-Lem 2022).

CHAT provides a social and cultural perspective on examining how institutional norms, historical practices, and individual agency intersect in the practice of musical collaboration (Burkitt 2021). It relates to piano accompaniment in terms of how cultural knowledge influences the performer's interpretation and contributes to the ensemble's collaborative work (Burner 2019; Leontiev 2020).

Combined, these constructs offer a comprehensive and integrated perspective on the interaction of technical, relational, and cultural dynamics in collaborative piano playing, providing the conceptual underpinning for this study's exploration of the lived experiences of musical practice among musicians.

Methodology

The research employed a qualitative methodology and thematic analysis framework to explore the intricate dynamics of piano accompaniment in collaborative music-making, analyzing musicians' experiences, the quality of available information, communication strategies, and cultural aspects. Conducted in China, a musically rich and culturally diverse setting, the study offers an ideal context for examining intercultural and interpretive collaboration.

Participants were selected through purposive sampling, resulting in a sample of 21 individuals, including professional musicians, educators, and advanced students involved in accompaniment. Data saturation was monitored and confirmed when no new themes emerged during the final interviews.

Semi-structured interviews elicited in-depth narratives of real-world experiences, focusing on collaborative interactions, emotional and informational exchange, and cultural influences. Interview questions were organized around four core themes and are briefly summarized in Table 1. The guide was pilot-tested and refined for clarity and neutrality.

To ensure reflexivity and minimize interpretive bias, the research team maintained positionality documentation, conducted regular peer debriefings, and used reflective memos throughout the process. Thematic analysis followed Braun and Clarke's six-phase framework, with transcripts coded inductively using NVivo 12. Inter-coder reliability was established through independent coding of 25% of transcripts by a second analyst, with discrepancies resolved through discussion. Measures were taken to reduce social desirability bias, including ensuring confidentiality and allowing participants to choose interview settings. Table 2 presents the demographic and professional details of the sample, demonstrating its diversity in terms of age, role, and musical background.

Ethical approval was obtained from the relevant institutional review board, and all participants provided informed consent prior to data collection. The study's qualitative design, combined with transparent methodological practices and diverse representation, enhances the credibility and transferability of findings within the field of collaborative music performance research.

Results

Understanding musicians' experiences. Participants described piano accompaniment as a multi-dimensional experience shaped by technical precision, emotional alignment, and interpersonal coordination. While they acknowledged the collaborative and artistic value of accompaniment, challenges such as tempo mismatches, interpretative conflicts, and dynamic imbalances frequently emerged. The accompanist's role, according to many, extends beyond technical support to include co-creation and interpretive leadership.

Rehearsals were repeatedly emphasized as crucial spaces for resolving artistic differences and building interpersonal trust. Interviewee 5 noted, "When I perform with an accompanist, I expect them to match my interpretative choices, but that alignment does not always come naturally. It often takes multiple rehearsals to achieve a cohesive sound." Similarly, Interviewee 12 stated, "A good accompanist listens actively and adjusts their playing to highlight the soloist's strengths while compensating for any weaknesses." These insights highlight accompaniment as an

Table 1 Interview questions and thematic domains.		
Theme	Sample questions	Objective
Understanding Musicians' Experiences	Can you describe a memorable experience where piano accompaniment played a significant role in your performance?	To explore participants' personal and artistic experiences with piano accompaniment and understand their interpretive and collaborative depth.
Understanding Musicians' Experiences	What challenges do you face during collaborative performances?	To identify common challenges in collaborative settings and examine how musicians navigate artistic coordination.
Assessing Information Quality	How do you ensure the information you receive about the piece or performance context is accurate?	To assess how musicians evaluate the reliability and usefulness of different information sources for performance preparation.
Assessing Information Quality	What sources of information do you rely on most when preparing for an accompaniment?	To understand the hierarchy of information sources and how musicians synthesize them in their preparatory process.
Exploring Communication Practices	How do you communicate your intentions and interpretations with your accompanist or soloist?	To explore the modes of verbal and nonverbal communication and their effect on ensemble cohesion.
Exploring Communication Practices	Can you share an instance where effective communication enhanced the performance outcome?	To illustrate how communication contributes to conflict resolution and performance synergy.
Identifying Cultural Influences	In what ways do you think cultural elements influence piano accompaniment practices in China?	To examine how cultural identity and tradition shape practices, expectations, and stylistic interpretation in accompaniment.
Identifying Cultural Influences	Have you experienced differences in accompaniment practices across cultural or stylistic contexts?	To understand the impact of cross-cultural exposure and stylistic diversity on musicians' adaptive strategies.

evolving partnership that requires shared artistic vision, adaptability, and mutual respect.

Three principal codes, *technical and emotional challenges, skill and relationship development*, and *holistic experience of accompaniment*, were distilled through coding stages, as summarized in Table 3 and visualized in Fig. 1. Table 3 outlines the progression of qualitative coding from open to axial and selective levels, illustrating how participants' experiences with technical challenges, emotional dynamics, and interpersonal coordination coalesce into a holistic understanding of piano accompaniment. Figure 1 illustrates the relative frequency of coded references related to synchronization challenges, emotional engagement, and technical skill development. It reflects how often each subtheme was emphasized across interviews, highlighting the multifaceted nature of musicians' experiences in collaborative piano accompaniment. These findings support an emphasis on interpersonal trust and advocacy for rehearsal as a foundation for relational and technical cohesion (Fullard and Cotter-Lockard 2025; Hughes et al. 2025).

Assessing information quality. Participants emphasized that the reliability, relevance, and clarity of information directly affect performance quality. While human sources, such as teachers and peers, were still favored for interpretive feedback, many also utilized AI tools for technical refinement. The tension between the emotional intuition of human feedback and the analytical precision of AI tools created a dynamic evaluation environment.

Interviewee 8 reflected, "AI tools help me analyze the technical aspects of my playing, like tempo consistency or dynamic range, but they lack the personal touch of advice from a mentor." Interviewee 17 added, "Sometimes, I receive contradictory suggestions from peers and teachers, which makes it hard to decide which direction to take".

This theme featured subcategories, including *the importance of personal sources, digital versus human feedback*, and *the dynamic role of information sources*, as shown in Table 4 and represented visually in Fig. 2. Table 4 presents emergent codes related to the sources, relevance, and accessibility of information. The coding layers highlight musicians' reliance on both human and digital feedback, emphasizing the importance of blended knowledge systems. Figure 2 illustrates how participants prioritized various sources of information, including teachers, AI tools, and peers, highlighting a preference for human guidance with contextual relevance. These results echo recommendations for culturally nuanced teaching and insights on the complementary role of AI and human feedback (Samuel et al. 2023; Yadav 2024).

Exploring communication practices. Communication, both verbal and nonverbal, emerged as a core element in collaborative music-making. Participants relied on real-time gestures, facial expressions, and eye contact to maintain synchrony during performances. In rehearsal contexts, verbal discussion played a vital role in resolving interpretative differences and aligning expectations.

Interviewee 3 shared, "During a performance, a simple nod or glance from the accompanist can make all the difference in staying synchronized." Conversely, Interviewee 11 stated, "Rehearsals are where we discuss our differences and ensure that we are on the same page. Without open dialogue, misunderstandings can carry over into the performance".

Three sub-themes were identified: *the importance of nonverbal cues, the balance between verbal and nonverbal cues, and the synchronization of communication*. These are presented in Table 5 and illustrated in Fig. 3. Table 3 captures participants' views on verbal and nonverbal communication during rehearsals and performances. The coding sequence illustrates how musicians strike a balance between explicit discussion and implicit signals to foster effective collaboration. Figure 3 represents the emphasis placed on verbal vs nonverbal strategies, with nonverbal cues receiving the most frequent coding references. The findings support the analysis of ensemble dynamics and argue for structured artistic communication in rehearsals (Paxton 2022).

Identifying cultural influences. Cultural background played a pivotal role in shaping musical expectations and interpretive decisions. Participants frequently mentioned that navigating between Western classical styles and traditional Chinese music required both technical sensitivity and cultural literacy. The process of cultural negotiation was viewed not only as a challenge but also as an opportunity for creating enriched performance experiences.

Interviewee 15 noted, "When we perform traditional Chinese pieces, the accompanist needs to understand the cultural essence behind the music to bring it to life." Similarly, Interviewee 20 remarked, "I have worked with accompanists who did not fully

Table 2 Participant demographics and professional details.							
Resp-ID	Age range	Primary role	Years of experience	Educational background	Musical focus	Main performance setting	City/region
R1	25-30	Accompanist	10	Bachelor's	Classical Piano	Solo Performances	Beijing
R2	30-35	Teacher	12	Master's	Western Classical	Classroom and Concerts	Shanghai
R3	35-40	Pianist	15	Bachelor's	Fusion, Classical	Orchestral and Solo Performances	Guangzhou
R4	40-45	Teacher	20	Doctorate	Classical and Contemporary	University and Concerts	Shenzhen
R5	28-33	Accompanist	9	Bachelor's	Classical Piano	Collaborative Performances	Hangzhou
R6	22-27	Performer	4	Undergraduate	Classical Piano	Recitals and University Performances	Nanjing
R7	45-50	Teacher	22	Doctorate	Western Classical	Concerts and Education	Beijing
R8	33-38	Accompanist	14	Master's	Classical Piano	Solo and Collaborative Performances	Chengdu
R9	25-30	Performer	3	Undergraduate	Classical Piano	Recitals and University Performances	Suzhou
R10	30-35	Pianist	11	Bachelor's	Classical Piano	Solo Performances	Xi'an
R11	38-43	Teacher	17	Master's	Classical and Jazz	University and Concerts	Shanghai
R12	40-45	Accompanist	18	Bachelor's	Classical Piano	Collaborative Performances	Shenzhen
R13	35-40	Teacher	16	Doctorate	Classical and Jazz	University and Concerts	Guangzhou
R14	27-32	Pianist	8	Bachelor's	Classical Piano	Recitals and Solo Performances	Hangzhou
R15	29-34	Performer	5	Undergraduate	Classical Piano	Recitals and University Performances	Beijing
R16	30-35	Teacher	15	Master's	Classical and Jazz	Concerts and Classroom	Chengdu
R17	33-38	Accompanist	13	Bachelor's	Classical Piano	Collaborative Performances	Suzhou
R18	22-27	Performer	3	Undergraduate	Classical Piano	University Performances	Nanjing
R19	40-45	Teacher	21	Doctorate	Classical and Contemporary	Concerts and Education	Beijing
R20	35-40	Pianist	16	Bachelor's	Classical Piano	Solo and Collaborative Performances	Xi'an
R21	28-33	Teacher	10	Master's	Classical and Jazz	University and Concerts	Shanghai

Table 3 Thematic coding structure for Theme 1.			
Step	Code	Description	Example from data
Step 1: Open Coding	Challenges in Synchronization	Difficulty in matching the timing and rhythm during performances.	"Sometimes, the timing is off, and it disrupts the entire performance." (Interviewee 1)
	Emotional Strain	Emotional discomfort due to the pressure of performance and collaboration.	"It's stressful when we don't connect emotionally during practice." (Interviewee 5)
	Technical Mastery	Need for refined technical skill to accompany soloists effectively.	"Being able to adapt to different musical styles is crucial." (Interviewee 10)
Step 2: Axial Coding	Technical and Emotional Challenges	Combination of technical difficulties and emotional pressures.	"It's not just about playing the right notes; it's about understanding the performer's mood." (Interviewee 8)
	Skill and Relationship Development	Developing both technical skills and interpersonal relationships with performers.	"The more we perform together, the better we understand each other's styles." (Interviewee 14)
Step 3: Selective Coding	Holistic Experience of Accompaniment	Both technical and emotional aspects shape the musician's experience.	"The experience is about balancing technical ability and emotional engagement." (Interviewee 3)

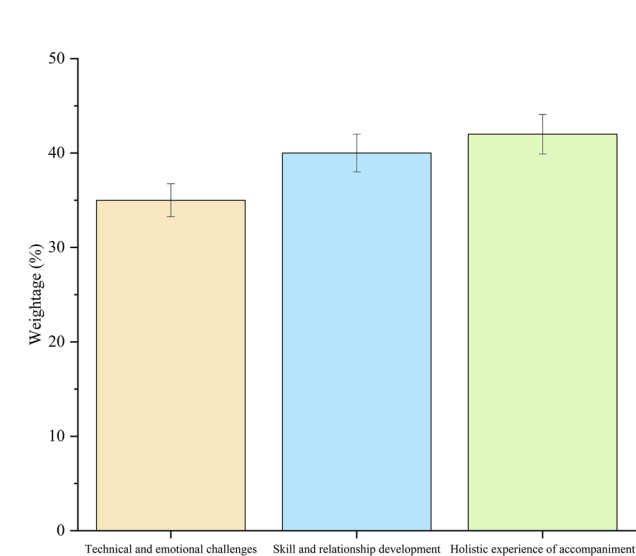


Fig. 1 Frequency-based coding distribution for Theme 1.

grasp the cultural context, which created a sense of disconnection in the music”.

From coding, three major themes emerged: *cultural sensitivity in collaborative music*, *East meets West in performance*, and *cultural integration in music performance*, detailed in Table 6 and depicted in Fig. 4. Table 6 outlines how cultural values, such as hierarchy, collectivism, and musical fusion, were coded from the interviews. The axial and selective coding phases highlight participants’ efforts to integrate Western and Chinese traditions in collaborative music-making. Figure 4 presents the relative coding density of cultural subthemes derived from participant interviews, including respect for hierarchy, blending Western and traditional Chinese musical styles, and collective performance values. The chart visualizes how often each cultural element was referenced, underscoring the significance of socio-cultural context in shaping accompaniment practices within Chinese musical traditions. These findings resonate with the discussion on cultural heritage in music interpretation and McMillan’s exploration of intercultural fusion in ensemble performance (Huang 2024).

Discussion

Piano accompaniment plays a central role in collaborative music-making, bringing together artistic expression, interpersonal coordination, and technical execution. This study examined this multifaceted phenomenon through the lived experiences of

musicians, focusing on four key dimensions: information quality, communication strategies, cultural influences, and collaborative dynamics. Drawing on qualitative data from professional musicians, educators, and advanced students in China, the discussion connects the study’s themes to existing academic literature, identifying new insights and outlining implications for musical practice, pedagogy, and future research.

Understanding musicians’ experiences. The results demonstrate that playing piano accompaniment is viewed by musicians as an artistic and relational practice, which requires them to strike a delicate balance between technical accuracy and emotional resonance. Differently put, we claim that successful accompaniment goes beyond support to involve interpretive work “in common” (Liu and Ye 2024). This is consistent with (Zhu 2021), who claims that being able to accompany well requires interpretive flexibility and emotional sensitivity. According to participants, the accompanist’s position shifts between mentorship in educational environments and co-creation in performance, highlighting the fluid and changing nature of musical partnerships.

These findings align with socio-constructivist perspectives on teaching and learning, such as the concept of scaffolding (Xu 2016), which posits that learning and artistic development are promoted through scaffolding collaboration in interaction with others. In addition, there appears to be a deficiency in role-specific accompanist training programs, thus emphasizing the expectation that accompanists’ education should address both the technical and interpersonal facets of their craft. This approach facilitates a more comprehensive understanding of the musician’s situation by integrating cognitive, emotional, and situational aspects.

Assessing information quality. Information quality and the information source were identified as core factors influencing the effectiveness of rehearsal and the level of confidence in performance. The role of mentorship and peer feedback was deeply appreciated by participants for its richness of context; however, they acknowledged the increasing utility of AI tools in assessing the technical aspects. This is evidence of the duality advocated by Information Quality Theory, which encompasses both accuracy and topical relevance (Shin et al. 2025).

For instance, one participant recalled receiving conflicting feedback: while their mentor emphasized expressive phrasing and emotional interpretation, an AI tool suggested flattening dynamic variations for technical consistency. This discrepancy caused hesitation in rehearsal choices, illustrating the cognitive

Table 4 Coding framework for Theme 2.			
Step	Code	Description	Example from data
Step 1: Open Coding	Relevance of Information	Quality of information in relation to its usefulness in practice.	"I need advice that's practical, not just theory." (Interviewee 2)
	Source of Information	Importance of where the information comes from—AI, educators, or peers.	"I prefer feedback from my teacher over online tutorials." (Interviewee 6)
	Information Accessibility	Ease of access to information that influences learning.	"Having access to online sheet music or AI-driven tutorials is very helpful." (Interviewee 9)
Step 2: Axial Coding	Importance of Personal Sources	The preference for personal, human sources of feedback over digital sources.	"There's something about the personal touch that you don't get from a machine." (Interviewee 15)
	Digital vs. Human Feedback	Comparison between digital tools and human feedback in terms of quality and reliability.	"AI feedback is good for technique, but it doesn't give you the emotional connection I need." (Interviewee 7)
Step 3: Selective Coding	Dynamic Role of Information Sources	Musicians value a blend of digital tools and human feedback for optimal performance.	"I use both: the speed of AI for technique, and the emotional guidance of my teacher." (Interviewee 18)

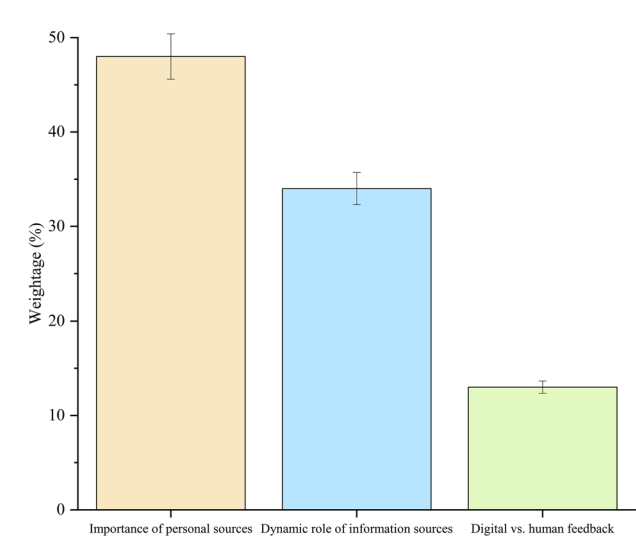


Fig. 2 Coding density for Theme 2.

negotiation musicians face when balancing algorithmic precision with human intuition.

The results also express ambivalent feelings towards digital resources as they struggle to convey emotional subtlety or cultural awareness. This is supported by (Shin et al. 2025), who argue that technology ought not to replace but supplement human intelligence in art. Participants called for a middle ground that combines old-school mentorship with modern tools, in other words, a hybrid artistic development suitable for 21st-century musicians.

Exploring communication practices. Communication became a logistical must and a creative impetus. Participants used non-verbal communication (such as gesture and eye contact) extensively in performance, favoring verbal (dialogue) in rehearsal to synchronize interpretive visions. Such findings are consistent with Collaborative Theory, which suggests that shared goals and mutual understanding are necessary for effective teamwork (Stienon and Tsaplina 2024).

Recent research further demonstrates that participatory inclusion and organizational responsiveness are key to creating psychological empowerment and cooperation in a dynamic team context (Bari et al. 2016a). These principles parallel the rehearsal context for ensemble instrumentalists, in which the level of the accompanist's agency, recognition, and interpretative voice contribute to communicative synergy. Interestingly, we found

nonverbal coordination and rehearsal conversation to be important mechanisms not only for artistic alignment but also for strengthening a sense of interpersonal 'equity' among performers.

The findings also indicate that both individual traits and ensemble environments shape how musicians communicate. For example, some shy participants reported difficulty expressing their artistic intentions, highlighting the need for training in expressive communication. In some cases, digital tools were used to address logistical barriers, especially in remote or long-distance collaborations. Overall, the results emphasize that effective communication in accompaniment involves a dynamic integration of verbal interaction, nonverbal coordination, and technological support.

Identifying cultural influences. Cultural context played a significant role in participants' feelings about and approach to accompaniment. Group-oriented Chinese cultural values (collectivism, respect for authority) and hierarchical social relations in ensemble playing were reflected in team dynamics, as ensemble players showed a preference for group value calculation over personal musical expression. This is consistent with theories such as the CHAT, which emphasizes that artistic practice is embedded in socio-cultural systems (Zhang and Leung 2024).

In culturally bound group work, psychological contracts and soft aspects, such as reciprocal expectations and participative management practices, significantly predict group results (Bari et al. 2016b). One interpretation of our results is that, in Chinese ensemble settings, accompanists operate within implicit socio-musical norms, negotiating interpretive independence while maintaining group coherence. This is indicative of the embeddedness of musical roles in larger cultural and human relations systems.

Participants described both the challenges and opportunities involved in blending Western classical techniques with traditional Chinese aesthetics, an effort that demands technical flexibility and cultural understanding. Younger musicians, in particular, expressed tension between the desire for modernization and the need to preserve cultural heritage. These findings highlight the importance of cultural literacy as a vital skill in piano accompaniment, enabling musicians to engage meaningfully with diverse artistic practices in collaborative settings.

Comparative analysis with previous studies. While much of what these findings have revealed resonates with qualitative studies to date, such as the roles of communication, interpretive flexibility, and emotional attunement in the accompaniment process, there are distinct points of departure. For example, in

Table 5 Thematic coding process for Theme 3.			
Step	Code	Description	Example from data
Step 1: Open Coding	Nonverbal Communication	Emphasis on gestures, facial expressions, and body language.	"I can tell if my accompanist is nervous by the way they're holding their hands." (Interviewee 4)
	Verbal Cues	Spoken communication regarding the interpretation of the piece.	"Before each performance, we always discuss the tempo and expression." (Interviewee 11)
	Mutual Understanding	Shared understanding developed over time through communication.	"We don't need to speak much anymore because we understand each other's playing style." (Interviewee 13)
Step 2: Axial Coding	Importance of Nonverbal Cues	Nonverbal cues take precedence in guiding musical expression and emotional interpretation.	"The way we look at each other is enough to understand the next phrase." (Interviewee 16)
	Verbal-Nonverbal Balance	Balance between verbal communication and nonverbal cues in rehearsals and performances.	"We talk less now, but we rely more on eye contact during performances." (Interviewee 12)
Step 3: Selective Coding	Synchronization of Verbal and Nonverbal Communication	Effective communication relies on both verbal and nonverbal practices, fostering smoother performances.	"It's a perfect mix of words and silent signals; that's what makes us click." (Interviewee 19)

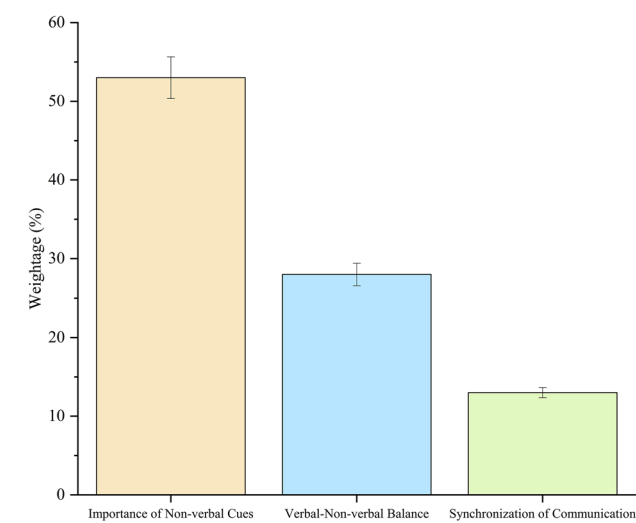


Fig. 3 Salience distribution of communication modalities in Theme 3.

comparison with Western investigations that emphasize personal expressiveness and autonomy in group contexts (He et al. 2024), this study demonstrates that these values are weakened and that there is a stronger inclination toward collectivism and hierarchical sensitivity (Liu and Ye 2024), influenced by traditional Chinese cultural norms. Participants often put their interpretation aside for the sake of group harmony, indicating that “accompaniment in China is not simply about a supportive role, but is a socially produced role, a role determined by social relational factors”. Furthermore, in contrast to studies in Euro-American contexts, where accompanists are commonly trained (Li 2018), this present study reveals a fascinating insight: There is limited formal training for accompanists in China, and much learning is based on informal modeling or learning from experience. These contrasting features emphasize that musical collaboration needs to be situated within its socio-cultural context and warrant regionally appropriate teaching paradigms that address the differences in practice and values.

Conclusion and implications. The focus here is on the entire experience of piano accompaniment, encompassing experiential, informational, communicative, and cultural aspects. Although it is outside the scope of this study, the thematic interplay identified

in this research confirms that good accompaniment is a success of the relationship and its context rather than solely a matter of personal skills. By situating these findings within existing theoretical and subject pedagogical literature, this discussion offers critical and practical insights.

Ultimately, this study advocates for a reevaluation of how piano accompaniment is taught. Effective pedagogy should focus not only on technical skills but also on practical insight, emotional awareness, information literacy, and cultural sensitivity. These recommendations have direct implications for curriculum development, ensemble training, and cross-cultural performance education. Future research could apply these insights to other forms of music-making and diverse cultural contexts to gain a deeper understanding of both shared and unique practices in collaborative performance. While the current findings are grounded in the Chinese context, they may offer a valuable point of reference for future cross-cultural or comparative studies, especially in societies where Western and indigenous music traditions coexist.

Conclusion

Study contribution and innovation. The novelty of this research lies in the merging of the cultural, informational, and communicative dimensions of piano accompaniment, particularly in a Chinese context. This framework has been missing in the literature. In contrast to earlier work that emphasizes either technical or social aspects in Western groups, this study foregrounds the cultural underpinnings of accompaniment. It presents a culturally grounded approach to analyzing joint music-making.

This paper presents a comprehensive examination of multiple perspectives on the dynamics of piano accompaniment in collaborative music-making, illustrating how musicians’ experiences and institutional, discursive, and cultural contexts shape their creative practices. In examining the Chinese musical landscape, this study establishes accompaniment as far more than just a technical task but as a socially embedded, emotionally charged, and culturally mediated activity. Results emphasize the importance of non-musical aspects of adaptive skills and suggest that a combined musical and social-emotional skill set is necessary for effective interpersonal relationships.

It offers novel perspectives related to music pedagogy and collaborative music-making by a) addressing the shortcomings of existing training and b) noticing the growing dependency on digital (and human) resources. It also raises the question of the

Table 6 Coding structure for Theme 4.			
Step	Code	Description	Example from data
Step 1: Open Coding	Respect for Hierarchy	Cultural respect for senior musicians and teachers.	"In China, you learn to respect your accompanist and follow their lead." (Interviewee 5)
	Blending Western and Traditional Music	Influence of Western classical and traditional Chinese music on playing styles.	"I have to incorporate the feeling of Chinese melodies while also staying true to classical structures." (Interviewee 14)
	Collective Approach to Music	Emphasis on group harmony over individual expression in Chinese musical culture.	"It's about the collective sound, not just one person standing out." (Interviewee 3)
Step 2: Axial Coding	Cultural Sensitivity in Collaborative Music	Sensitivity to the cultural background of performers enhances collaboration.	"When I play with other musicians from different regions of China, I adapt to their cultural approach." (Interviewee 11)
	East Meets West in Performance	The fusion of Western techniques with Chinese interpretive approaches to music-making.	"I feel the blend of Western structure with Chinese sensitivity makes our music richer." (Interviewee 7)
Step 3: Selective Coding	Cultural Integration in Music Performance	The effective blending of Western and traditional Chinese influences in creating harmonious collaborative performances.	"In the end, it's about respecting both traditions and combining them into a seamless performance." (Interviewee 17)

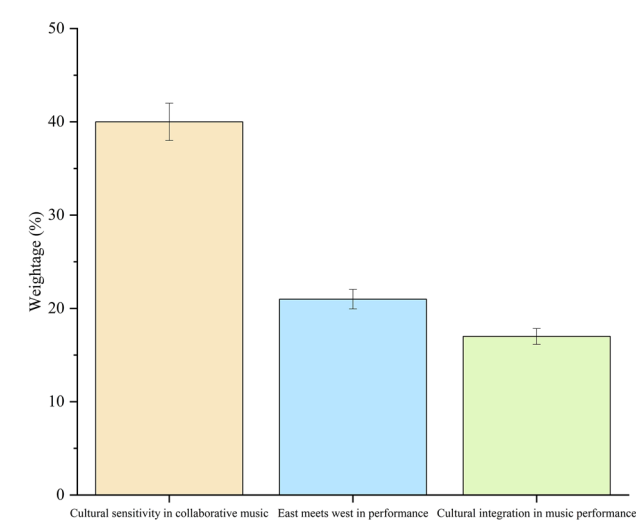


Fig. 4 Frequency distribution of codes related to Theme 4.

requirements of communication training and culturally sensitive education in an ensemble context. Through the adoption of theoretical perspectives, Collaboration Theory, Information Quality Theory, and Cultural-Historical Activity Theory, accompaniment is theorized here as a comprehensive approach to understanding skilled work in today's musical setting.

Implications for practice and pedagogy. The implications of these findings for music teachers, professional musicians, and schools are substantial. First, they emphasize the need for independent accompaniment training modules that focus on emotional intelligence, communication, and cultural fluency competencies, which are often lacking in most curricula. Second, a theme of expertise and peer learning emerged as a central element in interpretive development, and it is recommended that structured mentorship be more formally integrated into teaching programs.

In addition, professional ensemble musicians and performance presenters may need to be more aware of cultural influences and cooperative expectations. Particularly in transcultural and interdisciplinary contexts, cultural competence, awareness of one's own status, and nonverbal communication can contribute to ensemble cohesion and artistic integrity.

Theoretical contributions. This study contributes to multiple theoretical domains. It refines Collaboration Theory by showing how mutual understanding and adaptive communication are central to real-time musical alignment. It expands Information Quality Theory by applying it to the arts, showing that musicians evaluate information not only for accuracy but for emotional and interpretive relevance. Additionally, it extends Cultural Theory by demonstrating that accompaniment is inseparable from cultural values and historical practices, particularly in non-Western contexts such as China.

Limitations and directions for future research. While this research provides valuable insights, its scope is limited to the Chinese context, which may constrain the generalizability of findings. Future comparative studies should examine how accompaniment practices vary across different cultural and educational systems to validate and extend these conclusions. A cross-national design would help uncover both universal and context-specific patterns of collaboration.

Moreover, the reliance on interview data, although methodologically appropriate, may be enriched by mixed-methods designs that incorporate observational or performance-based data. Longitudinal studies could further explore how accompanists' roles evolve over time and in sustained collaborations.

The intersection of digital technology and traditional musical collaboration also invites deeper investigation. As AI-based tools and digital platforms become more integrated into music education and practice, future research should evaluate their impact on communication, emotional expression, and artistic authenticity in accompaniment. For example, studies on AI-human collaboration in ensemble performance could offer new pedagogical and technical frameworks for training accompanists.

Concluding remarks

In sum, this study foregrounds the complex, culturally embedded, and evolving nature of piano accompaniment in collaborative music-making. It challenges reductionist views of accompaniment as a subordinate or mechanical function, instead presenting it as a dynamic interplay of artistic vision, interpersonal negotiation, and contextual literacy. By proposing actionable pedagogical strategies and opening new theoretical inquiries, this research contributes to a more nuanced and globally aware understanding of music collaboration in the 21st century.

Data availability

The data supporting the findings of this study are included within the article. No additional datasets were generated or analyzed beyond those presented in the published manuscript.

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References

- Adhikari T (2023) MeloHarmony: exploring emotion in crafting AI-generated music with generative adversarial network powered harmony. <https://ssrn.com/abstract=4575257>
- Atanacković D (2024) Artificial intelligence: duality in applications of generative AI and assistive AI in music. *INSAM J Contemporary Music Art Technol* 12:12–31
- Atika ASL (2024) Interpersonal music communication in the Serenade Bunga Bangsa Music Group in the Song Anjayeng Bawana. *INJECT Interdiscip J Commun* 9(1):1–16
- Bari MW, Fanchen M, Baloch MA (2016a) Management practices and performance of mergers and acquisitions in Pakistan: mediating role of psychological contract. *SpringerPlus* 5(1):1527
- Bari MW, Fanchen M, Baloch MA (2016b) TQM soft practices and job satisfaction; mediating role of relational psychological contract. *Proc Soc Behav Sci* 235:453–462
- Ben-Tal O, Harris MT, Sturm BL (2021) How music AI is useful: engagements with composers, performers and audiences. *Leonardo* 54(5):510–516
- Burkitt I (2021) The emotions in cultural-historical activity theory: personality, emotion and motivation in social relations and activity. *Integr Psychol Behav Sci* 55(4):797–820
- Burner T (2019) Aligning assessment and learning using cultural-historical activity theory. *Assess Matters* 13:91–112
- Bussu A, Mangiarulo M (2024) Playing music together: exploring the impact of a classical music ensemble on adolescent's life skills self-perception. *PLoS One* 19(7):e0306326
- Clayton M (2016) The social and cultural function of music in cross-cultural perspective. In *The Oxford handbook of music psychology*, Oxford University Press, Oxford, United Kingdom, p. 47
- Cong-Lem N (2022) Unravelling cultural-historical activity theory (CHAT): Leontiev's and Engeström's approaches to activity theory. *Knowl Cult* 10(1):84–103
- Crane D (2016) Culture and globalization: Theoretical models and emerging trends. In: Crane, D., Kawashima, N. & Kawasaki, K. (eds.) *Global Culture: Media, Arts, Policy, and Globalization*. United Kingdom: Routledge, London, pp. 1–25
- Davis T (2023) Empathy and deep listening in jazz improvisation. In *The Oxford handbook of care in music education*, Oxford University Press, New York, NY, United States of America, pp. 205–216
- Foot KA (2014) Cultural-historical activity theory: exploring a theory to inform practice and research. *J Hum Behav Soc Environ* 24(3):329–347
- Fullard A, Cotter-Lockard D (2025) The art of collaboration: chamber music rehearsal techniques and team building. Oxford University Press, Oxford, United Kingdom
- Hall LM, Buechler C, Marusca G, Brennan S, Levine DL, Marusca GA, Brennan SK (2021) Utilizing chamber music to teach non-verbal communication to medical students: a pilot initiative. *Cureus* 13(4):e14587
- He M, Barkeshli M, Toroghi RM (2024) Cultural diplomacy in arts education and oil painting: a systematic review of east-west fusion artistic expressions. *Arts Educa*, 41, pp. 264–279
- Huang Y (2024) Cultural harmonies: exploring compositional techniques and cultural fusion in Guizhou Ethnic minority music. *Pac Int J* 7(1):216–221
- Hughes BT, Jacobson R, Rule NO, Srivastava S (2025) Stereotypes and social decisions: the interpersonal consequences of socioeconomic status. *J Person Soc Psychol* 128:1226–1242
- John R, John S, Azariah SK (2024) The policies of pedagogical accompaniment as a development strategy teaching professional: theoretical review. *Int J Soc Bus Sci* 2(1):64–76
- Jordanous A, Smaill A (2009) Investigating the role of score following in automatic musical accompaniment. *J N Music Res* 38(2):197–209
- Kubota Y (2009) The nature of professional accompanists and their roles: performing with musical excellence and enjoying communicative interaction. In *International Symposium on Performance Science*, European Association of Conservatoires (AEC), Utrecht, Netherlands
- Lee KJM, Pasquier P, Yuri J (2025) Revival: collaborative artistic creation through human-AI interactions in musical creativity. *Arxiv* <https://arxiv.org/abs/2503.15498>
- Leontiev DA (2020) Cultural-historical activity theory in the framework of the "functional paradigm. *Cultural Historical Psychol* 16(2):19–24
- Li S (2025) The impact of AI-driven music production software on the economics of the music industry. *Inf Dev* 41(3):957–967. <https://doi.org/10.1177/02666669241312170>
- Li Z (2018) Study on the important role of piano accompaniment in the performance of ethnic instruments. In *The Proceeding of the International Conference on Culture, Literature Arts & Humanities*, ICCLAH, Francis Academic Press, London, United Kingdom
- Liu B, Ye F (2024) The problem-based approach in online music education: how to teach students to control singing with piano accompaniment?. *Interact Learn Environ* 32(6):2862–2873
- Nugrahu PA, Sulistyowati R, Utami NNA, Ernest J (2023) Striving for musical excellence: a study on the development of music players' skills for church worship accompaniment through ensemble training. *Dewa Ruci. J Pengkaj Dan Pencipta Seni* 18(1):28–38
- Paxton J (2022) A new model of interpretation and communication for ensemble music-making. University of Sussex
- Puerto O, Thue D (2017) A Model of Inter-musician Communication for Artificial Musical Intelligence. In *ICCC*, pp. 221–228
- Rahman E, Webb WR, Rao P, Yu N, Garcia PE, Ioannidis S, Sayed K, Philipp-Dormston WG, Carruthers JD, Mosahebi A (2024) A systematic review on the reinforcement loop in aesthetic medicine and surgery: the interplay of social media, self-perception, and repeat procedures. *Aesthetic Plast Surg* 1:13
- Raza A (2025) The evolution of human-AI collaboration in creative industries: case studies in music and design. *Multidiscip Res Comput Inf Syst* 5(2):105–122
- Rieger T, Manzey D (2022) Human performance consequences of automated decision aids: the impact of time pressure. *Hum Factors* 64(4):617–634
- Royal Society (Great Britain). Study Group on Non-Verbal Communication (1972). *Non-verbal communication*. Cambridge University Press, Cambridge, England
- Saarikallio S (2012) Cross-cultural approaches to music and health. In: *Music, health, and wellbeing*. Oxford University Press, pp. 477–490
- Samuel Y, Brennan-Tonetta M, Samuel J, Kashyap R, Kumar V, Krishna Kaashyap S, Chidipothu N, Anand I, Jain P (2023) Cultivation of human centered artificial intelligence: culturally adaptive thinking in education (CATE) for AI. *Front Artif Intell* 6:1198180
- Seddon FA (2005) Modes of communication during jazz improvisation. *Br J Music Educ* 22(1):47–61
- Shin I, Hwang SB, Yoo YJ, Bae S, Kim RY (2025) Comparing student preferences for AI-generated and peer-generated feedback in AI-driven formative peer assessment. In *Proceedings of the 15th International Learning Analytics and Knowledge Conference*, Association for Computing Machinery (ACM), Dublin, Ireland
- Solomonova O, Ohanezova-Hryhorenko O, Demydova V, Kuchurivskiy Y, Bordonuk V (2023) Interpretive content of a musical work: the performing aspect. *Converg J Res Arts Educ* 16(32):125–138
- Springer DG, Silvey BA (2022) Effects of focus of attention instructions on listeners' evaluations of solo instrumental performance. *Int J Music Educ* 40(2):205–216
- Stienon Y, Tsaplina M (2024) Primitive way country come look inside. *Leonardo* 57(2):179–187
- Trehub SE, Becker J, Morley I (2015) Cross-cultural perspectives on music and musicality. *Philos Trans R Soc B Biol Sci* 370(1664):20140096
- Venkatesh S, Miranda ER, Braund E (2023) SSVEP-based brain-computer interface for music using a low-density EEG system. *Assist Technol* 35(5):378–388
- Wang X, Wei Y, Heng L, McAdams S (2021) A cross-cultural analysis of the influence of timbre on affect perception in western classical music and Chinese music traditions. *Front Psychol* 12:732865
- Waseem Bari M, Mahmood F, Bashir M (2021) The role of instrumental guanxi in the relation between entrepreneurs' social competence and firms' financial performance: a comparative study. *Econ Res Ekonomika istraživanja* 34(1):243–265
- Wilson V (2014) Examining teacher education through cultural-historical activity theory. *Teach Educ Adv Netw J* 6(1):20–29
- Wu K (2023) Cultural confluence: the impact of traditional and modern synergies in Chinese juvenile musical theater. *Int J Educ Humanit* 11(2):192–199
- Xu Z (2016) Current existing dilemma in harmonics teaching and counter-measures. In *4th International Conference on Management Science, Education Technology, Arts, Soc Sci Eco*, Atlantis Press, Paris, France
- Yadav D (2024) Navigating the landscape of ai integration in education: opportunities, challenges, and ethical considerations for harnessing the potential of artificial intelligence (AI) for teaching and learning. *BSSS J Comput* 15(1):38–48
- Zhan H (2025) Music education from a cross-cultural perspective: comparing music teaching methods in different countries. *SHS Web Conf* 213:02010
- Zhang C, Leung B-W (2024) Music majors' self-regulated learning (SRL) of music practice in China: two case studies based on the SRL microanalysis. *Int J Music Educ* 42(3):367–380
- Zhang J, Wang I (2024) Chinese and Western aesthetic realization in traditional Chinese instrumental performance: perspectives of scholars and students. *Malays J Soc Sci Humanit* 9(11):e003075–e003075

Zhu X (2021) Necessity and countermeasures of piano teaching reform for music majors in colleges and universities. In International Conference on Learning and Education Innovation, Atlantis Press, Paris, France, Vol. 26, pp. 21–23

Author contributions

YF conducted the preliminary research for this study, demonstrated its feasibility, and carried out the research. This specifically includes data collection, organization, analysis, and the writing and revision of the article.

Competing interests

The author declares no competing interests.

Ethics approval

All study procedures were approved by the Academic Committee of Leshan Normal University in March 2023 (Approval No. 1182021). The research protocols conformed to the ethical principles outlined in the Declaration of Helsinki. All participants were informed of the content of the study and were assured that it would cause no harm to them.

Informed consent

Written assent and informed consent were obtained from all participants and their parents or legal guardians. Informed consent time: September 2023. This is a non-interventional study and will not cause any harm to the participants. The anonymity of the participants will be guaranteed. All participants and their legal guardians have been informed of the reasons for conducting the study and the way the data will be used. There is no risk for all participants and this study is absolutely safe.

Additional information

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